Relations

Only One Option Correct

1. Let A be the set of all functions $f : Z \rightarrow Z$ and R be a relation on A such that $R = \{(f, g) : f(0) = g(1)$ and $f(1) = g(0)\}$. Then R is :
   (a) Transitive but neither reflexive nor symmetric
   (b) Symmetric and transitive but not reflexive
   (c) Symmetric but neither reflexive nor transitive
   (d) Reflexive but neither symmetric nor transitive.
   (Main 2nd April 1st Shift 2025)

2. Let $A = \{-3, -2, -1, 0, 1, 2, 3\}$. Let R be a relation on A defined by xRy if and only if $0 < x^2 + 2y \leq 4$. Let l be the number of elements in R and m be the minimum number of elements required to be added in R to make it a reflexive relation. Then $l + m$ is equal to
   (a) 18 (b) 20 (c) 17 (d) 19
   (Main 3rd April 1st Shift 2025)

3. Let $A = \{-2, -1, 0, 1, 2, 3\}$. Let R be a relation on A defined by xRy if and only if $y = \max\{x, 1\}$. Let l be the number of elements in R. Let m and n be the minimum number of elements required to be added in R to make it reflexive and symmetric relations, respectively. Then $l + m + n$ is equal to
   (a) 13 (b) 12 (c) 14 (d) 11
   (Main 3rd April 2nd Shift 2025)

4. Let $A = \{-3, -2, -1, 0, 1, 2, 3\}$ and R be a relation on A defined by xRy if and only if $2x - y \in \{0, 1\}$. Let l be the number of elements in R. Let m and n be the minimum number of elements required to be added in R to make it reflexive and symmetric relations, respectively. Then $l + m + n$ is equal to :
   (a) 17 (b) 16 (c) 18 (d) 15
   (Main 4th April 2nd Shift 2025)

5. The number of non-empty equivalence relations on the set $\{1, 2, 3\}$ is:
   (a) 5 (b) 6 (c) 7 (d) 4
   (Main 22nd Jan 1st Shift 2025)

6. Let $R = \{(1, 2), (2, 3), (3, 3)\}$ be a relation defined on the set $\{1, 2, 3, 4\}$. Then the minimum number of elements, needed to be added in R so that R becomes an equivalence relation, is:
   (a) 7 (b) 10 (c) 9 (d) 8
   (Main 23rd Jan 1st Shift 2025)

7. The relation $R = \{(x, y) : x, y \in Z$ and $x + y$ is even$\}$ is :
   (a) reflexive and transitive but not symmetric
   (b) reflexive and symmetric but not transitive
   (c) symmetric and transitive but not reflexive
   (d) an equivalence relation *(Main 28th Jan 1st Shift 2025)*

8. Define a relation R on the interval $\left[0, \frac{\pi}{2}\right)$ by $x\,R\,y$ if and only if $\sec^2 x - \tan^2 y = 1$. Then R is :
   (a) an equivalence relation
   (b) reflexive but neither symmetric not transitive
   (c) both reflexive and symmetric but not transitive
   (d) both reflexive and transitive but not symmetric
   (Main 29th Jan 1st Shift 2025)

9. Let the relations R_1 and R_2 on the set $X = \{1, 2, 3, ..., 20\}$ be given by $R_1 = \{(x, y): 2x - 3y = 2\}$ and $R_2 = \{(x, y) : -5x + 4y = 0\}$. If M and N be the minimum number of elements required to be added in R_1 and R_2, respectively, in order to make the relations symmetric, then $M+N$ equals
   (a) 10 (b) 8 (c) 16 (d) 12
   (Main 6th April 1st Shift 2024)

10. Let $A = \{1, 2, 3, 4, 5\}$. Let R be a relation on A defined by xRy if and only if $4x \leq 5y$. Let m be the number of elements in R and n be the minimum number of elements from $A \times A$ that are required to be added to R to make it a symmetric relation. Then $m + n$ is equal to
    (a) 24 (b) 26 (c) 25 (d) 23
    (Main 6th April 2nd Shift 2024)

11. Let $A = \{2, 3, 6, 8, 9, 11\}$ and $B = \{1, 4, 5, 10, 15\}$. Let R be a relation on $A \times B$ defined by $(a, b)\,R(c, d)$ if and only if $3ad - 7bc$ is an even integer. Then the relation R is
    (a) an equivalence relation.
    (b) reflexive and symmetric but not transitive.
    (c) reflexive but not symmetric.
    (d) transitive but not symmetric.
    (Main 8th April 2nd Shift 2024)

12. Consider the relations R_1 and R_2 defined as $aR_1b \Leftrightarrow a^2 + b^2 = 1$ for all $a, b \in \mathbf{R}$ and $(a, b)R_2(c, d) \Leftrightarrow a + d = b + c$ for all $(a, b), (c, d) \in \mathbf{N} \times \mathbf{N}$. Then
    (a) R_1 and R_2 both are equivalence relations
    (b) Only R_1 is an equivalence relation
    (c) Only R_2 is an equivalence relation
    (d) Neither R_1 nor R_2 is an equivalence relation
    (Main 1st Feb 2nd Shift 2024)

13. Let $S = \{1, 2, 3, . . ., 10\}$. Suppose M is the set of all the subsets of S, then the relation $R = \{(A, B) : A \cap B \neq \phi;$ $A, B \in M\}$ is
    (a) symmetric and transitive only
    (b) reflexive only
    (c) symmetric and reflexive only
    (d) symmetric only *(Main 27th Jan 1st Shift 2024)*

14. Let R be a relation on $Z \times Z$ defined by $(a, b)\,R(c, d)$ if and only if $ad - bc$ is divisible by 5. Then R is
    (a) Reflexive but neither symmetric nor transitive
    (b) Reflexive, symmetric and transitive
    (c) Reflexive and transitive but not symmetric
    (d) Reflexive and symmetric but not transitive
    (Main 29th Jan 1st Shift 2024)

15. If R is the smallest equivalence relation on the set $\{1, 2, 3, 4\}$ such that $\{(1, 2), (1, 3)\} \subset R$, then the number of elements in R is ________.
    (a) 12 (b) 15 (c) 10 (d) 8
    (Main 29th Jan 2nd Shift 2024)

16. Let $A = \{1, 2, 3, 4, 5, 6, 7\}$. Then the relation $R = \{(x, y) \in A \times A : x + y = 7\}$ is
(a) an equivalence relation
(b) reflexive but neither symmetric nor transitive
(c) transitive but neither symmetric nor reflexive
(d) symmetric but neither reflexive nor transitive
(Main 8th April 2nd Shift 2023)

17. Let **R** be a relation on **R**, given by
$\mathbf{R} = \{(a,b) : 3a - 3b + \sqrt{7}$ is an irrational number$\}$.
Then **R** is
(a) reflexive and symmetric but not transitive
(b) reflexive and transitive but not symmetric
(c) reflexive but neither symmetric nor transitive
(d) an equivalence relation *(Main 1st Feb 1st Shift 2023)*

18. Let $P(S)$ denote the power set of $S = \{1, 2, 3, ..., 10\}$. Define the relations R_1 and R_2 on $P(S)$ as AR_1B if $(A \cap B^C) \cup (B \cap A^C) = \phi$ and AR_2B if $A \cup B^C = B \cup A^C$, $\forall A, B \in P(S)$. Then
(a) both R_1 and R_2 are not equivalence relations
(b) only R_2 is an equivalence relation
(c) only R_1 is an equivalence relation
(d) both R_1 and R_2 are equivalence relations
(Main 1st Feb 2nd Shift 2023)

19. The relation $R = \{(a, b) : gcd(a, b) = 1, 2a \neq b, a, b, \in Z\}$ is
(a) reflexive but not symmetric
(b) transitive but not reflexive
(c) symmetric but not transitive
(d) neither symmetric nor transitive
(Main 24th Jan 1st Shift 2023)

20. Let R be a relation defined on $\mathbb{N}$ as $a R b$ if $2a + 3b$ is a multiple of 5, $a, b \in \mathbb{N}$.
(a) symmetric but not transitive
(b) not reflexive
(c) an equivalence relation
(d) transitive but not symmetric
(Main 29th Jan 2nd Shift 2023)

21. The minimum number of elements that must be added to the relation $R = \{(a, b), (b, c)\}$ on the set $\{a, b, c\}$ so that it becomes symmetric and transitive is
(a) 3 (b) 7 (c) 4 (d) 5
(Main 30th Jan 1st Shift 2023)

22. Let R be a relation on $N \times N$ defined by $(a, b) R(c, d)$ if and only if $ad(b - c) = bc(a - d)$. Then R is
(a) transitive but neither reflexive nor symmetric
(b) symmetric but neither reflexive nor transitive
(c) symmetric and transitive but not reflexive
(d) reflexive and symmetric but not transitive
(Main 31st Jan 1st Shift 2023)

23. Among the relations
$S = \{(a, b) : a, b \in \mathbf{R} - \{0\}, 2 + \frac{a}{b} > 0\}$ and
$T = \{(a, b): a, b \in \mathbf{R}, a^2 - b^2 \in Z\}$
(a) S is transitive but T is not
(b) both S and T are symmetric
(c) neither S nor T is transitive
(d) T is symmetric but S is not
(Main 31st Jan 2nd Shift 2023)

24. Let R_1 and R_2 be two relations defined on R by $aR_1b \Leftrightarrow ab \geq 0$ and $aR_2b \Leftrightarrow a \geq b$. Then,
(a) R_1 is an equivalence relation but not R_2
(b) R_2 is an equivalence relation but not R_1
(c) both R_1 and R_2 are equivalence relations
(d) neither R_1 nor R_2 is an equivalence relation
(Main 27th July 1st Shift 2022)

25. For $\alpha \in N$, consider a relation R on N given by $R = \{(x, y): 3x + \alpha y$ is a multiple of $7\}$. The relation R is an equivalence relation if and only if :
(a) $\alpha = 14$ (b) α is a multiple of 4
(c) 4 is the remainder when α is divided by 10
(d) 4 is the remainder when α is divided by 7
(Main 28th July 1st Shift 2022)

26. Let $R_1 = \{(a, b) \in N \times N : |a - b| \leq 13\}$ and $R_2 = \{(a, b) \in N \times N : |a - b| \neq 13\}$. Then on N:
(a) Both R_1 and R_2 are equivalence relations
(b) Neither R_1 nor R_2 is an equivalence relation
(c) R_1 is an equivalence relation but R_2 is not
(d) R_2 is an equivalence relation but R_1 is not
(Main 28th June 2nd Shift 2022)

27. Let a set $A = A_1 \cup A_2 \cup ...... \cup A_k$, where $A_i \cap A_j = \phi$ for $i \neq j$, $1 \leq i, j \leq k$. Define the relation R from A to A by $R = \{(x, y) : y \in A_i$ if and only if $x \in A_i, 1 \leq i \leq k\}$. Then R is
(a) reflexive, symmetric but not transitive
(b) reflexive, transitive but not symmetric
(c) reflexive but not symmetric and transitive
(d) an equivalence relation *(Main 29th June 1st Shift 2022)*

28. Which of the following is not correct for relation R on the set of real numbers ?
(a) $(x, y) \in R \Leftrightarrow |x| - |y| \leq 1$ is reflexive but not symmetric.
(b) $(x, y) \in R \Leftrightarrow |x - y| \leq 1$ is reflexive and symmetric.
(c) $(x, y) \in R \Leftrightarrow 0 < |x - y| \leq 1$ is symmetric and transitive.
(d) $(x, y) \in R \Leftrightarrow 0 < |x| - |y| \leq 1$ is neither transitive nor symmetric. *(Main 31st Aug 1st Shift 2021)*

29. Let N be the set of natural numbers and a relation R on N be defined by $R = \{(x, y) \in N \times N : x^3 - 3x^2y - xy^2 + 3y^3 = 0\}$. Then the relation R is
(a) reflexive but neither symmetric nor transitive
(b) an equivalence relation
(c) reflexive and symmetric, but not transitive
(d) symmetric but neither reflexive nor transitive
(Main 27th July 2nd Shift 2021)

30. Let $A = \{2, 3, 4, 5, ..., 30\}$ and '$\simeq$' be an equivalence relation on $A \times A$, defined by $(a, b) \simeq (c, d)$ if and only if $ad = bc$. Then the number of ordered pairs which satisfy this equivalence relation with ordered pair $(4, 3)$ is equal to
(a) 5 (b) 6 (c) 7 (d) 8
(Main 16th March 2nd Shift 2021)

31. Let $R = \{(P, Q) \mid P$ and Q are at the same distance from the origin$\}$ be a relation, then the equivalence class of $(1, -1)$ is the set
(a) $S = \{(x, y) \mid x^2 + y^2 = 1\}$
(b) $S = \{(x, y) \mid x^2 + y^2 = 2\}$
(c) $S = \{(x, y) \mid x^2 + y^2 = 4\}$
(d) $S = \{(x, y) \mid x^2 + y^2 = \sqrt{2}\}$
(Main 26th Feb 1st Shift 2021)

32. Let R_1 and R_2 be two relations defined as follows :
$R_1 = \{(a, b) \in R^2 : a^2 + b^2 \in Q\}$ and
$R_2 = \{(a, b) \in R^2 : a^2 + b^2 \notin Q\}$, where Q is the set of all rational numbers. Then
(a) Neither R_1 nor R_2 is transitive.
(b) R_2 is transitive but R_1 is not transitive.
(c) R_1 is transitive but R_2 is not transitive.
(d) R_1 and R_2 are both transitive.
(Main 3rd Sept 2nd Shift 2020)

33. Consider the following two binary relations on the set $A = \{a, b, c\}$:
$R_1 = \{(c, a), (b, b), (a, c), (c, c), (b, c), (a, a)\}$ and
$R_2 = \{(a, b), (b, a), (c, c), (c, a), (a, a), (b, b), (a, c)\}$.
Then
(a) R_1 is not symmetric but it is transitive
(b) both R_1 and R_2 are transitive
(c) R_2 is symmetric but it is not transitive
(d) both R_1 and R_2 are not symmetric *(Main Online 2018)*

34. Let N denote the set of all natural numbers. Define two binary relations on N as
$R_1 = \{(x, y) \in N \times N : 2x + y = 10\}$
and $R_2 = \{(x, y) \in N \times N : x + 2y = 10\}$. Then :
(a) Both R_1 and R_2 are symmetric relations
(b) Range of R_1 is $\{2, 4, 8\}$
(c) Both R_1 and R_2 are transitive relations
(d) Range of R_2 is $\{1, 2, 3, 4\}$ *(Main Online 2018)*

35. Consider the following relations:
$R = \{(x, y) | x, y$ are real numbers and $x = wy$ for some rational number $w\}$;
$S = \left\{\left(\frac{m}{n}, \frac{p}{q}\right) \middle| m, n, p \text{ and } q \text{ are integers such that } n, q \neq 0 \text{ and } qm = pn\right\}$. Then
(a) R is an equivalence relation but S is not an equivalence relation
(b) neither R nor S is an equivalence relation
(c) S is an equivalence relation but R is not an equivalence relation
(d) R and S both are equivalence relations *(AIEEE 2010)*

36. Let R be the real line. Consider the following subsets of the plane $R \times R$:
$S = \{(x, y) : y = x + 1 \text{ and } 0 < x < 2\}$
$T = \{(x, y) : x - y \text{ is an integer}\}$.
Which one of the following is true?
(a) T is an equivalence relation on R but S is not
(b) Neither S nor T is an equivalence relation on R
(c) Both S and T are equivalence relations on R
(d) S is an equivalence relation on R but T is not
(AIEEE 2008)

37. Let W denote the words in the English dictionary. Define the relation R by :
$R = \{(x, y) \in W \times W |$ the words x and y have at least one letter in common$\}$.
Then R is
(a) not reflexive, symmetric and transitive
(b) reflexive, symmetric and not transitive
(c) reflexive, symmetric and transitive
(d) reflexive, not symmetric and transitive.
(AIEEE 2006)

38. Let $R = \{(3, 3), (6, 6), (9, 9), (12, 12), (6, 12), (3, 9), (3, 12), (3, 6)\}$ be a relation on the set $A = \{3, 6, 9, 12\}$. The relation is
(a) reflexive and symmetric only
(b) an equivalence relation
(c) reflexive only
(d) reflexive and transitive only *(AIEEE 2005)*

39. Let $R = \{(1, 3), (4, 2), (2, 4), (2, 3), (3, 1)\}$ be a relation on the set $A = \{1, 2, 3, 4\}$. The relation R is
(a) not symmetric (b) transitive
(c) a function (d) reflexive *(AIEEE 2004)*

Numerical / Integer Value

40. The number of relations on the set $A = \{1, 2, 3\}$, containing at most 6 elements including (1, 2), which are reflexive and transitive but not symmetric, is ______ .
(Main 7th April 1st Shift 2025)

41. Let $A = \{1, 2, 3\}$. The number of relations on A, containing (1, 2) and (2, 3), which are reflexive and transitive but not symmetric, is ______ .
(Main 22nd Jan 2nd Shift 2025)

42. Let the set of all relations R on the set $\{a, b, c, d, e, f\}$, such that R is reflexive and symmetric, and R contains exactly 10 elements, be denoted by δ. Then the number of elements in δ is______. *(Advanced 2025)*

43. The number of symmetric relations defined on the set $\{1, 2, 3, 4\}$ which are not reflexive is ______.
(Main 30th Jan 2nd Shift 2024)

44. Let $A = \{1, 2, 3, 4\}$ and $R = \{(1, 2), (2, 3), (1, 4)\}$ be a relation on A. Let S be the equivalence relation on A such that $R \subset S$ and the number of elements in S is n. Then, the minimum value of n is ______.
(Main 31st Jan 1st Shift 2024)

45. Let $A = \{1, 2, 3, ..., 100\}$, Let R be a relation on A defined by $(x, y) \in R$ if and only if $2x = 3y$. Let R_1 be a symmetric relation on A such that $R \subset R_1$ and the number of elements in R_1 is n. Then, the minimum value of n is ______. *(Main 31st Jan 2nd Shift 2024)*

46. Let $A = \{0, 3, 4, 6, 7, 8, 9, 10\}$ and R be the relation defined on A such that $R = \{(x, y) \in A \times A : x - y$ is odd positive integer or $x - y = 2\}$. The minimum number of elements that must be added to the relation R, so that it is a symmetric relation, is equal to ______.
(Main 8th April 1st Shift 2023)

47. The number of relations, on the set $\{1, 2, 3\}$ containing (1, 2) and (2, 3), which are reflexive and transitive but not symmetric, is ______. *(Main 12th April 1st Shift 2023)*

48. Let $A = \{-4, -3, -2, 0\ 1, 3, 4\}$ and $R = \{(a, b) \in A \times A : b = |a| \text{ or } b^2 = a + 1\}$ be a relation on A. Then the minimum number of elements, that must be added to the relation R so that it becomes reflexive and symmetric, is ______
(Main 13th April 2nd Shift 2023)

49. The minimum number of elements that must be added to the relation $R = \{(a, b), (b, c), (b, d)\}$ on the set $\{a, b, c, d\}$ so that it is an equivalence relation is ______.
(Main 24th Jan 2nd Shift 2023)

Assertion and Reason / Statement Type

50. Let $A = \{0, 1, 2, 3, 4, 5\}$. Let R be a relation on A defined by $(x, y) \in R$ if and only if $\max\{x, y\} \in \{3, 4\}$. Then among the statements

(S_1) : The number of elements in R is 18, and

(S_2) : The relation R is symmetric but neither reflexive nor transitive.

(a) both are true
(b) only (S_2) is true
(c) both are false
(d) only (S_1) is true *(Main 8th April 2nd Shift 2025)*

51. Let $X = \mathbf{R} \times \mathbf{R}$. Define a relation R on X as:

$(a_1, b_1)\ R\ (a_2, b_2) \Leftrightarrow b_1 = b_2$.

Statement I : R is an equivalence relation.

Statement II : For some $(a, b) \in X$, the set $S = \{(x, y) \in X : (x, y)\ R\ (a, b)\}$ represents a line parallel to $y = x$.

In the light of the above statements, choose the correct answer from the options given below:

(a) Statement I is true but Statement II is false
(b) Statement I is false but Statement II is true
(c) Both Statement I and Statement II are false
(d) Both Statement I and Statement II are true

(Main 23rd Jan 2nd Shift 2025)

52. Let a relation R on $N \times N$ be defined as :

$(x_1, y_1)\ R\ (x_2, y_2)$ if and only if $x_1 \le x_2$ or $y_1 \le y_2$.

Consider the two statements :

(I) R is reflexive but not symmetric.
(II) R is transitive

Then which one of the following is true?

(a) Both (I) and (II) are correct.
(b) Neither (I) nor (II) is correct.
(c) Only (I) is correct.
(d) Only (II) is correct. *(Main 4th April 2nd Shift 2024)*

53. Let R be the set of real numbers.

Statement-1 : $A = \{(x, y) \in R \times R : y - x \text{ is an integer}\}$ is an equivalence relation on R.

Statement-2 : $B = \{(x, y) \in R \times R : x = \alpha y$ for some rational number $\alpha\}$ is an equivalence relation on R.

(a) Statement-1 is true, Statement-2 is false.
(b) Statement-1 is false, Statement-2 is true.
(c) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1.
(d) Statement-1 is true, Statement-2 is true; Statement-2 is not a correct explanation for Statement-1.

(AIEEE 2011)

Subjective Questions

54. A relation R on the set of complex numbers is defined by $z_1 R z_2$ if and only if $\dfrac{z_1 - z_2}{z_1 + z_2}$ is real show that R is an equivalence relation. *(IIT-JEE 1982)*

Functions

Only One Option Correct

55. If the domain of the function

$f(x) = \log_e\left(\dfrac{2x-3}{5+4x}\right) + \sin^{-1}\left(\dfrac{4+3x}{2-x}\right)$ is $[\alpha, \beta)$, then $\alpha^2 + 4\beta$ is equal to

(a) 7 (b) 5 (c) 3 (d) 4

(Main 3rd April 1st Shift 2025)

56. Let $f, g : (1, \infty) \to R$ be defined as $f(x) = \dfrac{2x+3}{5x+2}$ and $g(x) = \dfrac{2-3x}{1-x}$. If the range of the function $fog : [2, 4] \to R$ is $[\alpha, \beta]$, then $\dfrac{1}{\beta - \alpha}$ is equal to

(a) 56 (b) 68 (c) 29 (d) 2

(Main 4th April 1st Shift 2025)

57. Consider the sets $A = \{(x, y) \in R \times R : x^2 + y^2 = 25\}$ $B = \{(x, y) \in R \times R: x^2 + 9y^2 = 144\}$, $C = \{(x, y) \in Z \times Z : x^2 + y^2 \le 4\}$ and $D = A \cap B$. The total number of one -one functions from the set D to the set C is

(a) 18290 (b) 15120 (c) 17160 (d) 19320

(Main 4th April 1st Shift 2025)

58. Let $A = \{1, 2, 3, 4\}$ and $B = \{1, 4, 9, 16\}$. Then the number of many-one functions $f : A \to B$ such that $I \in f(A)$ is equal to :

(a) 163 (b) 139 (c) 151 (d) 127

(Main 22nd Jan 2nd Shift 2025)

59. Let $f(x) = \log_e x$ and $g(x) = \dfrac{x^4 - 2x^3 + 3x^2 - 2x + 2}{2x^2 - 2x + 1}$.

Then the domain of fog is

(a) $(0, \infty)$ (b) $[0, \infty)$ (c) $[1, \infty)$ (d) R

(Main 23rd Jan 1st Shift 2025)

60. The function $f : (-\infty, \infty) \to (-\infty, 1)$, defined by $f(x) = \dfrac{2^x - 2^{-x}}{2^x + 2^{-x}}$ is :

(a) Onto but not one-one
(b) Both one-one and onto
(c) One-one but not onto
(d) Neither one-one nor onto

(Main 24th Jan 2nd Shift 2025)

61. Let $f : [0, 3] \to A$ be defined by $f(x) = 2x^3 - 15x^2 + 36x + 7$ and $g : [0, \infty) \to B$ be defined by $g(x) = \dfrac{x^{2025}}{x^{2025} + 1}$. If both the functions are onto and $S = \{x \in \mathbf{Z}: x \in A \text{ or } x \in B\}$, then $n(S)$ is equal to :

(a) 36 (b) 29 (c) 30 (d) 31

(Main 28th Jan 2nd Shift 2025)

62. Let $f(x) = x^5 + 2e^{x/4}$ for all $x \in \mathbf{R}$. Consider a function $g(x)$ such that $(gof)(x) = x$ for all $x \in \mathbf{R}$. Then the value of $8g'(2)$ is

(a) 4 (b) 16 (c) 8 (d) 2

(Main 4th April 1st Shift 2024)

63. Let $A = \{1, 3, 7, 9, 11\}$ and $B = \{2, 4, 5, 7, 8, 10, 12\}$. Then the total number of one-one maps $f : A \to B$, such that $f(1) + f(3) = 14$, is

(a) 180 (b) 480 (c) 120 (d) 240

(Main 5th April 1st Shift 2024)

64. Let $f, g : R \to R$ be defined as :

$f(x) = |x - 1|$ and $g(x) = \begin{cases} e^x, & x \geq 0 \\ x+1, & x \leq 0 \end{cases}$.

Then the function $f(g(x))$ is

(a) onto but not one-one.

(b) neither one-one nor onto.

(c) both one-one and onto.

(d) one-one but not onto. *(Main 5th April 2nd Shift 2024)*

65. The function $f(x) = \dfrac{x^2+2x-15}{x^2-4x+9}, x \in R$ is

(a) both one-one and onto.

(b) neither one-one nor onto.

(c) onto but not one-one.

(d) one-one but not onto. *(Main 6th April 1st Shift 2024)*

66. Let $[t]$ be the greatest integer less than or equal to t. Let A be the set of all prime factors of 2310 and $f : A \to Z$ be the function $f(x) = \left[\log_2\left(x^2 + \left[\dfrac{x^3}{5}\right]\right)\right]$. The number of one-to-one functions from A to the range of f is

(a) 25 (b) 24 (c) 20 (d) 120

(Main 8th April 1st Shift 2024)

67. Let $f(x) = \begin{cases} -a \text{ if } -a \leq x \leq 0 \\ x + a \text{ if } 0 < x \leq a \end{cases}$

where $a > 0$ and $g(x) = (f(|x|) - |f(x)|)/2$.

Then the function $g : [-a, a] \to [-a, a]$ is

(a) onto.

(b) both one-one and onto.

(c) one-one.

(d) neither one-one nor onto.

(Main 8th April 2nd Shift 2024)

68. Let $f : R \to R$ and $g : R \to R$ be defined as

$f(x) = \begin{cases} \log_e x, & x > 0 \\ e^{-x}, & x \leq 0 \end{cases}$ and $g(x) = \begin{cases} x, & x \geq 0 \\ e^x, & x < 0 \end{cases}$.

Then, $gof : R \to R$ is

(a) neither one-one nor onto

(b) onto but not one-one

(c) one-one but not onto

(d) both one-one and onto *(Main 1st Feb 1st Shift 2024)*

69. Let $f(x) = \begin{cases} x-1, & x \text{ is even,} \\ 2x, & x \text{ is odd,} \end{cases}$ $x \in \mathbf{N}$. If for some $a \in \mathbf{N}$, $f(f(f(a))) = 21$, then $\lim\limits_{x \to a^-} \left\{\dfrac{|x|^3}{a} - \left[\dfrac{x}{a}\right]\right\}$, where $[t]$ denotes the greatest integer less than or equal to t, is equal to :

(a) 169 b) 121 (c) 225 (d) 144

(Main 1st Feb 2nd Shift 2024)

70. The function $f : N - \{1\} \to N$; defined by $f(n)$ = the highest prime factor of n, is :

(a) one-one only

(b) both one-one and onto

(c) neither one-one nor onto

(d) onto only *(Main 27th Jan 1st Shift 2024)*

71. Let $f : R - \left\{\dfrac{-1}{2}\right\} \to R$ and $g : R - \left\{\dfrac{-5}{2}\right\} \to R$ be defined as $f(x) = \dfrac{2x+3}{2x+1}$ and $g(x) = \dfrac{|x|+1}{2x+5}$. Then, the domain of the function fog is

(a) $R - \left\{-\dfrac{5}{2}, -\dfrac{7}{4}\right\}$ (b) $R - \left\{-\dfrac{7}{4}\right\}$

(c) R (d) $R - \left\{-\dfrac{5}{2}\right\}$

(Main 27th Jan 2nd Shift 2024)

72. If $f(x) = \begin{cases} 2+2x, & -1 \leq x < 0 \\ 1 - \dfrac{x}{3}, & 0 \leq x \leq 3 \end{cases}$; $g(x) = \begin{cases} -x, & -3 \leq x \leq 0 \\ x, & 0 < x \leq 1 \end{cases}$, then range of $(fog)(x)$ is

(a) [0, 1) (b) [0, 3) (c) (0, 1] (d) [0, 1]

(Main 29th Jan 1st Shift 2024)

73. If $f(x) = \dfrac{4x+3}{6x-4}, x \neq \dfrac{2}{3}$ and $(fof)(x) = g(x)$, where $g : R - \left\{\dfrac{2}{3}\right\} \to R - \left\{\dfrac{2}{3}\right\}$, then $(gogog)(4)$ is equal to

(a) $\dfrac{19}{20}$ (b) $-\dfrac{19}{20}$ (c) –4 (d) 4

(Main 31st Jan 1st Shift 2024)

74. If the function $f : (-\infty, -1\,] \to (a, b]$ defined by $f(x) = e^{x^3-3x+1}$ is one - one and onto, then the distance of the point $P(2b + 4, a + 2)$ from the line $x + e^{-3}y = 4$ is

(a) $2\sqrt{1+e^6}$ (b) $4\sqrt{1+e^6}$

(c) $3\sqrt{1+e^6}$ (d) $\sqrt{1+e^6}$

(Main 31st Jan 2nd Shift 2024)

75. If $f(x) = \dfrac{(\tan 1°)x + \log_e(123)}{x \log_e(1234) - (\tan 1°)}, x > 0$, then the least value of $f(f(x)) + f\left(f\left(\dfrac{4}{x}\right)\right)$ is

(a) 4 (b) 2 (c) 0 (d) 8

(Main 10th April 1st Shift 2023)

76. For $x \in \mathbb{R}$, two real valued functions $f(x)$ and $g(x)$ are such that, $g(x) = \sqrt{x} + 1$ and $fog(x) = x + 3 - \sqrt{x}$. Then $f(0)$ is equal to

(a) 5 (b) 1 (c) 0 (d) –3

(Main 13th April 1st Shift 2023)

77. Let $f : \mathbf{R} \to \mathbf{R}$ be a function such that

$f(x) = \dfrac{x^2+2x+1}{x^2+1}$. Then

(a) $f(x)$ is many-one in $(-\infty, -1)$
(b) $f(x)$ is one-one in $(-\infty, \infty)$
(c) $f(x)$ is many-one in $(1, \infty)$
(d) $f(x)$ is one-one in $[1, \infty)$ but not in $(-\infty, \infty)$

(Main 29th Jan 1st Shift 2023)

78. Let $f, g : N - \{1\} \to N$ be functions defined by $f(a) = \alpha$, where α is the maximum of the powers of those primes p such that p^{α} divides a, and $g(a) = a + 1$, for all $a \in N - \{1\}$. Then, the function $f + g$ is
(a) one-one but not onto
(b) onto but not one-one
(c) both one-one and onto
(d) neither one-one nor onto

(Main 27th July 1st Shift 2022)

79. The domain of the function
$$f(x)=\sin^{-1}[2x^2-3]+\log_2\left(\log_{1/2}(x^2-5x+5)\right),$$
where $[t]$ is the greatest integer function, is
(a) $\left(-\frac{\sqrt{5}}{2}, \frac{5-\sqrt{5}}{2}\right)$ (b) $\left(\frac{5-\sqrt{5}}{2}, \frac{5+\sqrt{5}}{2}\right)$
(c) $\left(1, \frac{5-\sqrt{5}}{2}\right)$ (d) $\left[1, \frac{5+\sqrt{5}}{2}\right)$

(Main 27th July 2nd Shift 2022)

80. The domain of the function $f(x)=\dfrac{\cos^{-1}\left(\dfrac{x^2-5x+6}{x^2-9}\right)}{\log_e(x^2-3x+2)}$ is
(a) $(-\infty, 1) \cup (2, \infty)$
(b) $(2, \infty)$
(c) $[-1/2,1)\cup(2,\infty)$
(d) $[-1/2,1)\cup(2,\infty)-\left\{\frac{3+\sqrt{5}}{2},\frac{3-\sqrt{5}}{2}\right\}$

(Main 24th June 1st Shift 2022)

81. Let $f(x)=\dfrac{x-1}{x+1}, x\in R-\{0,-1,1\}$.
If $f^{n+1}(x) = f(f^n(x))$ for all $n \in N$, then $f^6(6) + f^7(7)$ is equal to :
(a) $\frac{7}{6}$ (b) $-\frac{3}{2}$ (c) $\frac{7}{12}$ (d) $-\frac{11}{12}$

(Main 26th June 1st Shift 2022)

82. Let $f: R \to R$ be defined as $f(x) = x - 1$ and $g: R - \{1, -1\} \to R$ be defined as $g(x)=\dfrac{x^2}{x^2-1}$. Then the function fog is :
(a) one-one but not onto
(b) onto but not one-one
(c) both one-one and onto
(d) neither one-one nor onto

(Main 26th June 2nd Shift 2022)

83. Let a function $f: N \to N$ be defined by
$$f(n)=\begin{cases} 2n, & n=2,4,6,8,\ldots \\ n-1, & n=3,7,11,15,\ldots \\ \dfrac{n+1}{2}, & n=1,5,9,13,\ldots \end{cases}$$
then f is
(a) one-one but not onto
(b) onto but not one-one
(c) neither one-one nor onto
(d) one-one and onto

(Main 28th June 1st Shift 2022)

84. Let $f:R-\left\{\frac{\alpha}{6}\right\}\to R$ be defined by $f(x)=\dfrac{5x+3}{6x-\alpha}$. Then the value of α for which $(fof)(x) = x$, for all $x\in R-\left\{\frac{\alpha}{6}\right\}$, is
(a) 6 (b) 8
(c) 5 (d) No such α exists.

(Main 20th July 2nd Shift 2021)

85. Let $g: N \to N$ be defined as
$g(3n+1) = 3n + 2$, $g(3n + 2) = 3n + 3$,
$g(3n + 3) = 3n + 1$, for all $n \geq 0$.
Then which of the following statements is true?
(a) There exists a function $f: N \to N$ such that $gof = f$
(b) There exists a one-one function $f: N \to N$ such that $fog = f$
(c) There exists an onto function $f: N \to N$ such that $fog = f$
(d) $gogog = g$

(Main 25th July 1st Shift 2021)

86. Consider functions $f: A \to B$ and $g: B \to C$ $(A, B, C \subseteq R)$ such that $(gof)^{-1}$ exists, then
(a) f is one-one and g is onto
(b) f and g both are onto
(c) f is onto and g is one-one
(d) f and g both are one-one

(Main 25th July 2nd Shift 2021)

87. The inverse of $y = 5^{\log x}$ is
(a) $x=y^{\frac{1}{\log 5}}$ (b) $x=5^{\frac{1}{\log y}}$
(c) $x=y^{\log 5}$ (d) $x=5^{\log y}$

(Main 17th March 1st Shift 2021)

88. Let $f: R - \{3\} \to R - \{1\}$ be defined by $f(x)=\dfrac{x-2}{x-3}$.
Let $g: R \to R$ be given as $g(x) = 2x - 3$. Then, the sum of all the values of x for which $f^{-1}(x)+g^{-1}(x)=\frac{13}{2}$, is equal to
(a) 2 (b) 7 (c) 5 (d) 3

(Main 18th March 2nd Shift 2021)

89. Let $f : R \to R$ be defined as $f(x) = 2x - 1$ and $g : R - \{1\} \to R$ be defined as $g(x)=\dfrac{x-\frac{1}{2}}{x-1}$. Then the composition function $f(g(x))$ is
(a) onto but not one-one
(b) neither one-one nor onto
(c) one-one but not onto
(d) both one-one and onto

(Main 24th Feb 1st Shift 2021)

90. Let $f, g : N \to N$ such that $f(n + 1) = f(n) + f(1)\ \forall\ n \in N$ and g be any arbitrary function. Which of the following statements is not true?

(a) If f is onto, then $f(n) = n\ \forall\ n \in N$
(b) f is one-one
(c) If fog is one-one, then g is one-one.
(d) If g is onto, then fog is one-one.
(Main 25th Feb 1st Shift 2021)

91. Let x denote the total number of one-one functions from a set A with 3 elements to a set B with 5 elements and y denote the total number of one-one functions from the set A to the set $A \times B$. Then
(a) $y = 273x$ (b) $2y = 91x$
(c) $y = 91x$ (d) $2y = 273x$
(Main 25th Feb 2nd Shift 2021)

92. Let $A = \{1, 2, 3, ..., 10\}$ and $f: A \to A$ be defined as
$$f(k)=\begin{cases} k+1 \text{ if } k \text{ is odd} \\ k \quad \text{if } k \text{ is even} \end{cases}$$
Then the number of possible functions $g: A \to A$ such that $gof = f$ is
(a) 5^5 (b) ${}^{10}C_5$ (c) $5!$ (d) 10^5
(Main 26th Feb 2nd Shift 2021)

93. For a suitably chosen real constant a, let a function, $f: R - \{-a\} \to R$ be defined by $f(x)=\dfrac{a-x}{a+x}$. Further suppose that for any real number $x \neq -a$ and $f(x) \neq -a$, $(fof)(x) = (x)$. Then $f\left(-\dfrac{1}{2}\right)$ is equal to
(a) 1/3 (b) −1/3 (c) −3 (d) 3
(Main 6th Sept 2nd Shift 2020)

94. If $g(x) = x^2 + x - 1$ and $(gof)(x) = 4x^2 - 10x + 5$, then $f\left(\dfrac{5}{4}\right)$ is equal to
(a) $-\dfrac{3}{2}$ (b) $-\dfrac{1}{2}$ (c) $\dfrac{1}{2}$ (d) $\dfrac{3}{2}$
(Main 7th Jan 1st Shift 2020)

95. The inverse function of $f(x)=\dfrac{8^{2x}-8^{-2x}}{8^{2x}+8^{-2x}}$, $x \in (-1, 1)$, is ________.
(a) $\dfrac{1}{4}(\log_8 e)\log_e\left(\dfrac{1-x}{1+x}\right)$
(b) $\dfrac{1}{4}(\log_8 e)\log_e\left(\dfrac{1+x}{1-x}\right)$
(c) $\dfrac{1}{4}\log_e\left(\dfrac{1+x}{1-x}\right)$ (d) $\dfrac{1}{4}\log_e\left(\dfrac{1-x}{1+x}\right)$
(Main 8th Jan 1st Shift 2020)

96. Let $f: (1, 3) \to R$ be a function defined by $f(x)=\dfrac{x[x]}{1+x^2}$, where $[x]$ denotes the greatest integer $\leq x$. Then the range of f is
(a) $\left(\dfrac{2}{5},\dfrac{3}{5}\right]\cup\left(\dfrac{3}{4},\dfrac{4}{5}\right)$ (b) $\left(\dfrac{2}{5},\dfrac{4}{5}\right]$
(c) $\left(\dfrac{3}{5},\dfrac{4}{5}\right)$ (d) $\left(\dfrac{2}{5},\dfrac{1}{2}\right)\cup\left(\dfrac{3}{5},\dfrac{4}{5}\right]$
(Main 8th Jan 2nd Shift 2020)

97. If the function $f: R \to R$ is defined by $f(x) = |x|(x - \sin x)$, then which of the following statements is TRUE?
(a) f is one-one, but NOT onto
(b) f is onto, but NOT one-one
(c) f is BOTH one-one and onto
(d) f is NEITHER one-one NOR onto *(Advanced 2020)*

98. If the function $f: R-\{1,-1\} \to A$ defined by $f(x)=\dfrac{x^2}{1-x^2}$, is surjective, then A is equal to :
(a) $[0, \infty)$ (b) $R - (-1, 0)$
(c) $R - \{-1\}$ (d) $R - [-1, 0)$
(Main 9th April 1st Shift 2019)

99. Let $f(x) = x^2$, $x \in R$. For any $A \subseteq R$, define $g(A) = \{x \in R : f(x) \in A\}$. If $S = [0, 4]$, then which one of the following statements is not true?
(a) $g(f(S)) = g(S)$ (b) $g(f(S)) \neq S$
(c) $f(g(S)) \neq f(S)$ (d) $f(g(S)) = S$
(Main 10th April 1st Shift 2019)

100. Let $f(x) = \log_e(\sin x)$, $(0 < x < \pi)$ and $g(x) = \sin^{-1}(e^{-x})$, $(x \geq 0)$. If α is a positive real number such that $a = (fog)'(\alpha)$ and $b = (fog)(\alpha)$, then
(a) $a\alpha^2 + b\alpha + a = 0$ (b) $a\alpha^2 - b\alpha - a = 0$
(c) $a\alpha^2 + b\alpha - a = -2\alpha^2$ (d) $a\alpha^2 - b\alpha - a = 1$
(Main 10th April 2nd Shift 2019)

101. For $x \in (0, 3/2)$, let $f(x)=\sqrt{x}$, $g(x) = \tan x$ and $h(x)=\dfrac{1-x^2}{1+x^2}$. If $\phi(x) = ((hof)og)(x)$, then $\phi(\pi/3)$ is equal to
(a) $\tan\dfrac{\pi}{12}$ (b) $\tan\dfrac{5\pi}{12}$ (c) $\tan\dfrac{7\pi}{12}$ (d) $\tan\dfrac{11\pi}{12}$
(Main 12th April 1st Shift 2019)

102. For $x \in R - \{0, 1\}$, let $f_1(x)=\dfrac{1}{x}$, $f_2(x)=1-x$ and $f_3(x)=\dfrac{1}{1-x}$ be three given functions. If a function, $J(x)$ satisfies $(f_2oJof_1)(x) = f_3(x)$ then $J(x)$ is equal to
(a) $f_2(x)$ (b) $\dfrac{1}{x}f_3(x)$ (c) $f_3(x)$ (d) $f_1(x)$
(Main 9th Jan 1st Shift 2019)

103. Let $A = \{x \in R : x$ is not a positive integer$\}$. Define a function $f: A \to R$ as $f(x)=\dfrac{2x}{x-1}$, then f is
(a) surjective but not injective
(b) injective but not surjective
(c) not injective
(d) neither injective nor surjective
(Main 9th Jan 2nd Shift 2019)

104. Let N be the set of natural numbers and two functions f and g be defined as $f, g: N \to N$ such that
$$f(n)=\begin{cases} \dfrac{n+1}{2}, & \text{if } n \text{ is odd} \\ \dfrac{n}{2}, & \text{if } n \text{ is even} \end{cases}$$ and $g(n) = n - (-1)^n$.
Then fog is

(a) both one-one and onto
(b) one-one but not onto
(c) neither one-one nor onto
(d) onto but not one-one *(Main 10th Jan 2nd Shift 2019)*

105. Let a function $f: (0, \infty) \rightarrow (0, \infty)$ be defined by

$f(x) = \left|1 - \frac{1}{x}\right|$. Then f is

(a) injective only
(b) both injective as well as surjective
(c) not injective but it is surjective
(d) neither injective nor surjective

(Main 11th Jan 2nd Shift 2019)

106. Let $f : A \rightarrow B$ be a function defined as $f(x) = \frac{x-1}{x-2}$, where $A = R - \{2\}$ and $B = R - \{1\}$. Then f is

(a) Invertible and $f^{-1}(y) = \frac{2y-1}{y-1}$
(b) Not invertible
(c) Invertible and $f^{-1}(y) = \frac{3y-1}{y-1}$
(d) Invertible and $f^{-1}(y) = \frac{2y+1}{y-1}$ *(Main Online 2018)*

107. The function $f: R \rightarrow \left[-\frac{1}{2}, \frac{1}{2}\right]$ defined as

$f(x) = \frac{x}{1+x^2}$, is

(a) injective but not surjective
(b) surjective but not injective
(c) neither injective nor surjective
(d) invertible *(Main 2017)*

108. Let $f(x) = 2^{10}\cdot x + 1$ and $g(x) = 3^{10}\cdot x - 1$. If $(fog)(x) = x$, then x is equal to

(a) $\frac{3^{10}-1}{3^{10}-2^{-10}}$ (b) $\frac{2^{10}-1}{2^{10}-3^{-10}}$

(c) $\frac{1-3^{-10}}{2^{10}-3^{-10}}$ (d) $\frac{1-2^{-10}}{3^{10}-2^{-10}}$

(Main Online 2017)

109. The function $f : N \rightarrow N$ defined by $f(x) = x - 5\left[\frac{x}{5}\right]$, where N is the set of natural numbers and $[x]$ denotes the greatest integer less than or equal to x, is

(a) one-one and onto.
(b) onto but not one-one.
(c) neither one-one nor onto.
(d) one-one but not onto. *(Main Online 2017)*

110. For $x \in R$, $x \neq 0$, $x \neq 1$, let $f_0(x) = \frac{1}{1-x}$ and $f_{n+1}(x) = f_0(f_n(x))$, $n = 0, 1, 2$ .... Then the value of $f_{100}(3) + f_1\left(\frac{2}{3}\right) + f_2\left(\frac{3}{2}\right)$ is equal to

(a) $\frac{8}{3}$ (b) $\frac{4}{3}$ (c) $\frac{5}{3}$ (d) $\frac{1}{3}$

(Main Online 2016)

111. The function $f: [0, 3] \rightarrow [1, 29]$, defined by $f(x) = 2x^3 - 15x^2 + 36x + 1$, is

(a) one-one and onto (b) onto but not one-one
(c) one-one but not onto (d) neither one-one nor onto

(IIT-JEE 2012)

112. Let $f(x) = x^2$ and $g(x) = \sin x$ for all $x \in R$. Then the set of all x satisfying $(fogogof)(x) = (gogof)(x)$, where $(fog)(x) = f(g(x))$, is

(a) $\pm\sqrt{n\pi}, n \in \{0, 1, 2, ...\}$
(b) $\pm\sqrt{n\pi}, n \in \{1, 2, ...\}$
(c) $\frac{\pi}{2} + 2n\pi, n \in \{..., -2, -1, 0, 1, 2, ...\}$
(d) $2n\pi$, $n \in \{..., -2, -1, 0, 1, 2, ...\}$ *(IIT-JEE 2011)*

113. For real x, let $f(x) = x^3 + 5x + 1$, then

(a) f is onto on R but not one-one
(b) f is one-one and onto on R
(c) f is neither one-one nor onto on R
(d) f is one-one but not onto on R *(AIEEE 2009)*

114. Let $f: N \rightarrow Y$ be a function defined as $f(x) = 4x + 3$ where $Y = \{y \in N : y = 4x + 3$ for some $x \in N\}$. Show that f is invertible and its inverse is

(a) $g(y) = \frac{y-3}{4}$ (b) $g(y) = \frac{3y+4}{3}$

(c) $g(y) = 4 + \frac{y+3}{4}$ (d) $g(y) = \frac{y+3}{4}$ *(AIEEE 2008)*

115. If $F(x) = \left(f\left(\frac{x}{2}\right)\right)^2 + \left(g\left(\frac{x}{2}\right)\right)^2$ where $f''(x) = -f(x)$ and $g(x) = f'(x)$ and given that $F(5) = 5$, then $F(10)$ is equal to

(a) 5 (b) 10 (c) 0 (d) 15

(IIT-JEE 2006)

116. Let $f: (-1, 1) \rightarrow B$, be a function defined by

$f(x) = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$, then f is both one-one and onto when B is the interval

(a) $\left[0, \frac{\pi}{2}\right)$ (b) $\left(0, \frac{\pi}{2}\right)$

(c) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (d) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ *(AIEEE 2005)*

117. $f(x) = \begin{cases} x, & \text{if } x \text{ is rational} \\ 0, & \text{if } x \text{ is irrational} \end{cases}$,

and $g(x) = \begin{cases} 0, & \text{if } x \text{ is rational} \\ x, & \text{if } x \text{ is irrational} \end{cases}$. Then $f - g$ is

(a) neither one-one nor onto
(b) one-one and onto
(c) one-one and into
(d) many one and onto *(IIT-JEE 2005)*

118. X and Y are two sets and $f: X \rightarrow Y$. If $\{f(c) = y; c \subset X, y \subset Y\}$, then the true statement is

(a) $f^{-1}(f(a)) = a$ (b) $f(f^{-1}(b)) = b, b \subset Y$
(c) $f^{-1}(f(a)) = a, a \subset X$ (d) $f(f^{-1}(b)) = b$

(IIT-JEE 2005)

119. If $f: R \to S$, defined by $f(x) = \sin x - \sqrt{3}\cos x + 1$, is onto, then the interval of S is
(a) [0, 1] (b) [−1, 1]
(c) [0, 3] (d) [−1, 3] *(AIEEE 2004)*

120. The graph of the function $y = f(x)$ is symmetrical about the line $x = 2$, then
(a) $f(x) = f(-x)$ (b) $f(2 + x) = f(2 - x)$
(c) $f(x + 2) = f(x - 2)$ (d) $f(x) = -f(-x)$ *(AIEEE 2004)*

121. The domain of the function $f(x) = \dfrac{\sin^{-1}(x-3)}{\sqrt{9-x^2}}$ is
(a) [1, 2] (b) [2, 3)
(c) [2, 3] (d) [1, 2) *(AIEEE 2004)*

122. If $f(x) = \sin x + \cos x$, $g(x) = x^2 - 1$, then $g(f(x))$ is invertible in the domain
(a) $\left[0, \dfrac{\pi}{2}\right]$ (b) $\left[-\dfrac{\pi}{4}, \dfrac{\pi}{4}\right]$
(c) $\left[-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right]$ (d) $[0, \pi]$ *(IIT-JEE 2004)*

123. A function f from the set of natural numbers to integers defined by $f(n) = \begin{cases} \dfrac{n-1}{2}, & \text{when } n \text{ is odd} \\ -\dfrac{n}{2}, & \text{when } n \text{ is even} \end{cases}$ is
(a) onto but not one-one
(b) one-one and onto both
(c) neither one-one nor onto
(d) one-one but not onto *(AIEEE 2003)*

124. If $f: [0, \infty) \to [0, \infty)$, and $f(x) = \dfrac{x}{1+x}$, then f is
(a) one-one and onto
(b) one-one but not onto
(c) onto but not one-one
(d) neither one-one nor onto *(IIT-JEE 2003)*

125. Domain of definition of the function $f(x) = \sqrt{\sin^{-1}(2x) + \dfrac{\pi}{6}}$ for real valued x, is
(a) $\left[-\dfrac{1}{4}, \dfrac{1}{2}\right]$ (b) $\left[-\dfrac{1}{2}, \dfrac{1}{2}\right]$
(c) $\left(-\dfrac{1}{2}, \dfrac{1}{9}\right)$ (d) $\left[-\dfrac{1}{4}, \dfrac{1}{4}\right]$ *(IIT-JEE 2003)*

126. The domain of $\sin^{-1}\left[\log_3\left(\dfrac{x}{3}\right)\right]$ is
(a) [1, 9] (b) [−1, 9]
(c) [−9, 1] (d) [−9, −1] *(AIEEE 2002)*

127. Let function $f: R \to R$ be defined by $f(x) = 2x + \sin x$ for $x \in R$. Then f is
(a) one-to-one and onto
(b) one-to-one but NOT onto
(c) onto but NOT one-to-one
(d) neither one-to-one nor onto *(IIT-JEE 2002)*

128. The domain of the derivative of the function
$f(x) = \begin{cases} \tan^{-1} x, & \text{if } |x| \le 1 \\ \dfrac{1}{2}(|x|-1), & \text{if } |x| > 1 \end{cases}$ is
(a) $R - \{0\}$ (b) $R - \{1\}$
(c) $R - \{-1\}$ (d) $R - \{-1, 1\}$ *(IIT-JEE 2002)*

129. Let $E = \{1, 2, 3, 4\}$ and $F = \{1, 2\}$, then the number of onto functions from E to F is
(a) 14 (b) 16 (c) 12 (d) 8 *(IIT-JEE 2001)*

130. If $f: [1, \infty) \to [2, \infty)$ is given by $f(x) = x + \dfrac{1}{x}$, then $f^{-1}(x)$ equals
(a) $\dfrac{x+\sqrt{x^2-4}}{2}$ (b) $\dfrac{x}{1+x^2}$
(c) $\dfrac{x-\sqrt{x^2-4}}{4}$ (d) $1+\sqrt{x^2-4}$ *(IIT-JEE 2001)*

131. Let $g(x) = 1 + x - [x]$ and $f(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0 \end{cases}$ then for all x, $f(g(x))$ is equal to
(a) x (b) 1
(c) $f(x)$ (d) $g(x)$ *(IIT-JEE 2001)*

132. Let $f(x) = \dfrac{\alpha x}{x+1}$, $x \ne -1$, then for what value of α is $f(f(x)) = x$?
(a) $\sqrt{2}$ (b) $-\sqrt{2}$
(c) 1 (d) −1 *(IIT-JEE 2001)*

133. Let $f: R \to R$ be any function. Defined $g: R \to R$ by $g(x) = |f(x)|$ for all x. Then g is
(a) onto if f is onto
(b) one-one if f is one-one
(c) continuous if f is continuous
(d) differentiable if f is differentiable *(IIT-JEE 2000)*

134. If the function $f: [1, \infty) \to [1, \infty)$ is defined by $f(x) = 2^{x(x-1)}$, then $f^{-1}(x)$ is
(a) $\left(-\dfrac{1}{2}\right)^{x(x-1)}$ (b) $\dfrac{1}{2}\left(1+\sqrt{1+4\log_2 x}\right)$
(c) $\dfrac{1}{2}\left(1-\sqrt{1+4\log_2 x}\right)$ (d) none of these *(IIT-JEE 1999)*

135. If $f(x) = \dfrac{x^2-1}{x^2+1}$, for every real number x, then the minimum value of f
(a) does not exist because f is unbounded
(b) is not attained even though f is bounded
(c) is equal to 1
(d) is equal to −1 *(IIT-JEE 1998)*

136. If $f(x) = 3x - 5$, then $f^{-1}(x)$
(a) is given by $\dfrac{1}{3x-5}$ (b) is given by $\dfrac{x+5}{3}$
(c) does not exist because f is not one - one
(d) does not exist because f is not onto. *(IIT-JEE 1998)*

137. If $g(f(x)) = |\sin x|$ and $f(g(x)) = (\sin\sqrt{x})^2$, then

(a) $f(x)=\sin^2 x, g(x)=\sqrt{x}$

(b) $f(x) = \sin x, g(x) = |x|$

(c) $f(x)=x^2, g(x)=\sin\sqrt{x}$

(d) f and g cannot be determined *(IIT-JEE 1998)*

138. The number of values of x where the function $f(x)=\cos x + \cos(\sqrt{2}x)$ attains its maximum, is

(a) 0 (b) 1

(c) 2 (d) infinite *(IIT-JEE 1998)*

139. Let $f(x) = (x+1)^2 - 1, x \geq -1$. Then the set $\{x : f(x) = f^{-1}(x)\}$ is

(a) $\left\{0, -1, \dfrac{-3+i\sqrt{3}}{2}, \dfrac{-3-i\sqrt{3}}{2}\right\}$

(b) $\{0, 1, -1\}$ (c) $\{0, -1\}$

(d) ϕ *(IIT-JEE 1995)*

140. The function $f(x) = |px - q| + r|x|, x \in (-\infty, \infty)$ where $p > 0$, $q > 0, r > 0$ assumes its minimum value only on one point if

(a) $p \neq q$ (b) $r \neq q$

(c) $r \neq p$ (d) $p = q = r$ *(IIT-JEE 1995)*

141. Let $f(x)$ be defined for all $x > 0$ and be continuous.

If $f(x)$ satisfy $f\left(\dfrac{x}{y}\right) = f(x) - f(y)$ for all x, y and $f(e) = 1$, then

(a) $f(x)$ is bounded (b) $f\left(\dfrac{1}{x}\right) \to 0$ as $x \to 0$

(c) $x f(x) \to 1$ as $x \to 0$ (d) $f(x) = \ln x$ *(IIT-JEE 1995)*

142. Let $f(x) = \sin x$ and $g(x) = \ln|x|$. If the ranges of the composition functions fog and gof are R_1 and R_2 respectively, then

(a) $R_1 = \{u : -1 \leq u < 1\}, R_2 = \{v : -\infty < v < 0\}$

(b) $R_1 = \{u : -\infty < u < 0\}, R_2 = \{v : -1 \leq v \leq 0\}$

(c) $R_1 = \{u : -1 < u < 1\}, R_2 = \{v : -\infty < v < 0\}$

(d) $R_1 = \{u : -1 \leq u \leq 1\}, R_2 = \{v : -\infty < v \leq 0\}$

(IIT-JEE 1994)

One or More Than One Option(s) Correct

143. Let $\mathbb{N}$ denote the set of all natural numbers, and $\mathbb{Z}$ denote the set of all integers. Consider the functions $f : \mathbb{N} \to \mathbb{Z}$ and $g : \mathbb{Z} \to \mathbb{N}$ defined by

$$f(n) = \begin{cases} (n+1)/2 & \text{if } n \text{ is odd,} \\ (4-n)/2 & \text{if } n \text{ is even,} \end{cases}$$

and $g(n) = \begin{cases} 3+2n & \text{if } n \geq 0 \\ -2n & \text{if } n < 0 \end{cases}$.

Define $(gof)(n) = g(f(n))$ for all $n \in \mathbb{N}$, and $(fog)(n) = f(g(n))$ for all $n \in \mathbb{Z}$.

Then which of the following statements is (are) TRUE?

(a) gof is NOT one-one and gof is NOT onto

(b) fog is NOT one-one and fog is onto

(c) g is one-one and g is onto

(d) f is NOT one-one but f is onto *(Advanced 2025)*

144. Let $f : R \to R, g : R \to R$ and $h : R \to R$ be differentiable functions such that $f(x) = x^3 + 3x + 2$, $g(f(x)) = x$ and $h(g(g(x))) = x$ for all $x \in R$. Then

(a) $g'(2) = \dfrac{1}{15}$ (b) $h'(1) = 666$

(c) $h(0) = 16$ (d) $h(g(3)) = 36$

(Advanced 2016)

145. Let $f(x) = \sin\left(\dfrac{\pi}{6}\sin\left(\dfrac{\pi}{2}\sin x\right)\right)$ for all $x \in R$ and $g(x) = \dfrac{\pi}{2}\sin x$ for all $x \in R$. Let $(fog)(x)$ denote $f(g(x))$ and $(gof)(x)$ denote $g(f(x))$. Then which of the following is (are) true?

(a) Range of f is $\left[-\dfrac{1}{2}, \dfrac{1}{2}\right]$

(b) Range of fog is $\left[-\dfrac{1}{2}, \dfrac{1}{2}\right]$ (c) $\lim\limits_{x\to 0} \dfrac{f(x)}{g(x)} = \dfrac{\pi}{6}$

(d) There is an $x \in R$ such that $(gof)(x) = 1$

(Advanced 2015)

146. Let $f : (0, 1) \to R$ be defined by $f(x) = \dfrac{b-x}{1-bx}$, where b is a constant such that $0 < b < 1$. Then

(a) f is not invertible on $(0, 1)$

(b) $f \neq f^{-1}$ on $(0, 1)$ and $f'(b) = \dfrac{1}{f'(0)}$

(c) $f = f^{-1}$ on $(0, 1)$ and $f'(b) = \dfrac{1}{f'(0)}$

(d) f^{-1} is differentiable on $(0, 1)$ *(IIT-JEE 2011)*

147. $y = f(x) = \dfrac{x+2}{x-1}$, then

(a) $x = f(y)$

(b) $f(1) = 3$

(c) y increases with x for $x < 1$

(d) f is a rational function of x. *(IIT-JEE 1984)*

Numerical / Integer Value

148. Consider the function $f : R \to R$ defined by $f(x) = \dfrac{2x}{\sqrt{1+9x^2}}$. If the composition of $\underbrace{f,(f\,o\,f\,o\,f\,o \ldots\ldots o\,f)}_{10\text{ times}}(x) = \dfrac{2^{10}x}{\sqrt{1+9\alpha x^2}}$, then the value of $\sqrt{3\alpha+1}$ is equal to ______. *(Main 4th April 2nd Shift 2024)*

149. Let $A = \{(x, y) : 2x + 3y = 23, x, y \in N\}$ and $B = \{x : (x, y) \in A\}$. Then the number of one-one functions from A to B is equal to ______.

(Main 9th April 2nd Shift 2024)

150. let $A = \{1, 2, 3, \ldots, 7\}$ and let $P(A)$ denote the power set of A. If the number of functions $f : A \to P(A)$ such that $a \in f(a), \forall\, a \in A$ is m^n, m and $n \in N$ and m is least, then $m + n$ is equal to ______. *(Main 30th Jan 1st Shift 2024)*

151. Let $R = \{a, b, c, d, e\}$ and $S = \{1, 2, 3, 4\}$. Total number of onto functions $f : R \to S$ such that $f(a) \neq 1$, is equal to ______. *(Main 8th April 2nd Shift 2023)*

152. For some $a, b, c \in N$, let $f(x) = ax - 3$ and $g(x) = x^b + c$, $x \in R$. If $(fog)^{-1}(x) = \left(\frac{x-7}{2}\right)^{1/3}$, then $(fog)(ac) + (gof)(b)$ is equal to ______. *(Main 25th Jan 1st Shift 2023)*

153. Let $S = \{1, 2, 3, 4, 5, 6\}$. Then the number of one-one functions $f : S \to P(S)$, where $P(S)$ denote the power set of S, such that $f(n) \subset f(m)$ where $n < m$ is ______. *(Main 30th Jan 1st Shift 2023)*

154. Let $f^1(x) = \frac{3x+2}{2x+3}, x \in R - \left\{\frac{-3}{2}\right\}$

For $n \geq 2$, define $f^n(x) = f^1 o f^{n-1}(x)$.

If $f^5(x) = \frac{ax+b}{bx+a}, \gcd(a,b) = 1$, then $a + b$ is equal to ______. *(Main 30th Jan 1st Shift 2023)*

155. The number of one-one function $f : \{a, b, c, d\} \to \{0, 1, 2, ..., 10\}$ such that $2f(a) - f(b) + 3f(c) + f(d) = 0$ is ______. *(Main 24th June 1st Shift 2022)*

156. Let $f : R \to R$ be a function defined by

$$f(x) = \left(2\left(1 - \frac{x^{25}}{2}\right)(2 + x^{25})\right)^{1/50}.$$

If the function $g(x) = f(f(f(x))) + f(f(x))$, then the greatest integer less than or equal to $g(1)$ is ______.

(Main 25th June 1st Shift 2022)

157. Let $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Define $f : S \to S$ as

$$f(n) = \begin{cases} 2n, & \text{if } n = 1, 2, 3, 4, 5 \\ 2n - 11, & \text{if } n = 6, 7, 8, 9, 10 \end{cases}$$

Let $g : S \to S$ be a function such that

$$fog(n) = \begin{cases} n+1, & \text{if } n \text{ is odd} \\ n-1, & \text{if } n \text{ is even} \end{cases}$$

Then $g(10)(g(1) + g(2) + g(3) + g(4) + g(5))$ is equal to ______. *(Main 27th June 2nd Shift 2022)*

158. Let $S = \{1, 2, 3, 4\}$. Then the number of elements in the set $\{f : S \times S \to S : f$ is onto and $f(a, b) = f(b, a) \geq a$ $\forall (a, b) \in S \times S\}$ is ______.

(Main 28th June 2nd Shift 2022)

159. Let $f(x)$ and $g(x)$ be two real polynomials of degree 2 and 1 respectively. If $f(g(x)) = 8x^2 - 2x$, and $g(f(x)) = 4x^2 + 6x + 1$, then the value of $f(2) + g(2)$ is ______. *(Main 29th June 2nd Shift 2022)*

160. Let $A = \{0, 1, 2, 3, 4, 5, 6, 7\}$. Then the number of bijective functions $f : A \to A$ such that $f(1) + f(2) = 3 - f(3)$ is equal to ______. *(Main 22nd July 2nd Shift 2021)*

161. Let $A = \{a, b, c\}$ and $B = \{1, 2, 3, 4\}$. Then the number of elements in the set $C = \{f : A \to B \mid 2 \in f(A)$ and f is not one-one$\}$ is ______. *(Main 5th Sept 2nd Shift 2020)*

162. Let X be a set with exactly 5 elements and Y be a set with exactly 7 elements. If α is the number of one-one functions from X to Y and β is the number of onto functions from Y to X, then the value of $\frac{1}{5!}(\beta - \alpha)$ is ______. *(Advanced 2018)*

163. For any real number x, let $[x]$ denote the largest integer less than or equal to x. Let f be a real valued function defined on the interval $[-10, 10]$ by

$$f(x) = \begin{cases} x - [x], & \text{if } [x] \text{ is odd} \\ 1 + [x] - x, & \text{if } [x] \text{ is even} \end{cases}$$

Then the value of $\frac{\pi^2}{10}\int_{-10}^{10} f(x)\cos \pi x \, dx$ is ______.

(IIT-JEE 2010)

Matching List

164. Let $E_1 = \left\{x \in \mathbb{R} : x \neq 1 \text{ and } \frac{x}{x-1} > 0\right\}$ and

$E_2 = \left\{x \in E_1 : \sin^{-1}\left(\log_e\left(\frac{x}{x-1}\right)\right) \text{ is a real number}\right\}$.

(Here, the inverse trigonometric function $\sin^{-1}x$ assumes values in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.)

Let $f : E_1 \to \mathbb{R}$ be the function defined by

$$f(x) = \log_e\left(\frac{x}{x-1}\right)$$

and $g : E_2 \to \mathbb{R}$ be the function defined by

$$g(x) = \sin^{-1}\left(\log_e\left(\frac{x}{x-1}\right)\right).$$

	List-I		List-II
P.	The range of f is	1.	$\left(-\infty, \frac{1}{1-e}\right] \cup \left[\frac{e}{e-1}, \infty\right)$
Q.	The range of g contains	2.	$(0, 1)$
R.	The domain of f contains	3.	$\left[-\frac{1}{2}, \frac{1}{2}\right]$
S.	The domain of g is	4.	$(-\infty, 0) \cup (1, \infty)$
		5.	$\left(-\infty, \frac{e}{e-1}\right]$
		6.	$(-\infty, 0) \cup \left(\frac{1}{2}, \frac{e}{e-1}\right]$

The correct option is :

(a) P → 4; Q → 2; R → 1; S → 1
(b) P → 3; Q → 3; R → 6; S → 5
(c) P → 4; Q → 2; R → 1; S → 6
(d) P → 4; Q → 3; R → 6; S → 5 *(Advanced 2018)*

165. Let the functions defined in column I have domain $\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$ and range $(-\infty, \infty)$.

Column I	Column II
(i) $1+2x$	(A) onto but not one-one
(ii) $\tan x$	(B) one to one but not onto
	(C) one one and onto
	(D) neither one-one nor onto

(IIT-JEE 1992)

Comprehension

Paragraph for Q.(166 and 167) :

Let $S = \{1, 2, 3, 4, 5, 6\}$ and X be the set of all relations R from S to S that satisfy both the following properties :

i. R has exactly 6 elements.

ii. For each $(a, b) \in R$, we have $|a - b| \geq 2$.

Let $Y = \{R \in X : \text{The range of } R \text{ has exactly one element}\}$ and $Z = \{R \in X : R \text{ is a function from } S \text{ to } S\}$.

Let $n(A)$ denote the number of elements in a set A.

166. If $n(X) = {}^mC_6$, then the value of m is __________.

167. If the value of $n(Y) + n(Z)$ is k^2, then $|k|$ is __________.

(Advanced 2024)

Assertion and Reason / Statement Type

168. Let $f(x) = (x+1)^2 - 1, x \geq -1$.

Statement-1 : The set $\{x : f(x) = f^{-1}(x)\} = \{0, -1\}$.

Statement-2 : f is bijection.

(a) Statement-1 is true, Statement-2 is false.

(b) Statement-1 is false, Statement-2 is true.

(c) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1.

(d) Statement-1 is true, Statement-2 is true; Statement-2 is not a correct explanation for Statement-1

(AIEEE 2009)

Subjective Questions

169. A function $f : R \to R$, where R is the set of real numbers, is defined by $f(x) = \dfrac{\alpha x^2 + 6x - 8}{\alpha + 6x - 8x^2}$. Find the interval of values of α for which f is onto. Is the function one-to-one for $\alpha = 3$? Justify your answer. *(IIT-JEE 1996)*

170. Let f be a one-one function with domain $\{x, y, z\}$ and range $\{1, 2, 3\}$. It is given that exactly one of the following statements is true and the remaining statements are false.

$f(x) = 1, f(y) \neq 1, f(z) \neq 2$.

Determine $f^{-1}(1)$. *(IIT-JEE 1982)*

171. Let $f(x + y) = f(x)\, f(y)$ for all x and y. Suppose $f(5) = 2$ and $f'(0) = 3$. Find $f'(5)$. *(IIT-JEE 1981)*

172. Given $A = \left\{x : \frac{\pi}{6} \leq x \leq \frac{\pi}{3}\right\}$ and $f(x) = \cos x - x(x+1)$. Find $f(A)$. *(IIT-JEE 1980)*

True or False

173. If $f(x) = (a - x^n)^{1/n}$ where $a > 0$ and n is a positive integer, then $f[f(x)] = x$. *(IIT-JEE 1983)*

174. The function $f(x) = \dfrac{x^2 + 4x + 30}{x^2 - 8x + 18}$ is not one-to-one. *(IIT-JEE 1983)*

Fill in the Blanks

175. If $f(x) = \sin^2 x + \sin^2\left(x + \frac{\pi}{3}\right) + \cos x \cos\left(x + \frac{\pi}{3}\right)$ and $g\left(\frac{5}{4}\right) = 1$, then $(gof)(x) = ...$ *(IIT-JEE 1996)*

176. Let A be a set of n distinct elements. Then the total number of distinct functions from A to A is .... and out of these .... are onto functions. *(IIT-JEE 1985)*

177. The domain of the function $f(x) = \sin^{-1}\left(\log_2 \frac{x^2}{2}\right)$ is given by.................. *(IIT-JEE 1984)*

178. The values of $f(x) = 3 \sin\left(\sqrt{\frac{\pi^2}{16} - x^2}\right)$ lie in the interval........... *(IIT-JEE 1983)*